

Codetantra Practical 3

Practice Lab Assignment

The screenshot displays the Codetantra web application interface. The browser address bar shows the URL: `mitaoe.codetantra.com/secure/course.jsp?euclid=698c54860aba96214c8b7d5e#/contents/698c54860aba96214c8b7d65/698c54860aba96214c8b7d66/6684d40e5d95376955a73c1d`. The application header includes the Codetantra logo, navigation links (Home, Learn Anywhere), and user information (202501090056@mitaoe.ac.in, Support, Logout).

The main content area is titled "Essentials of Data Science Laboratory - 2304102L - 2025". A sidebar on the left lists the course structure, with "3.1.1. NumPy Array Operations" selected. The main panel displays the assignment details for "3.1.1. NumPy Array Operations".

Assignment Instructions:

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`.
- The shape of the array using `shape`.
- The total number of elements in the array using `size`.

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases:

Test case 1

Input	Output
3 5	[[1 2 3 4 5]]
5 4 7 8	[[1 2 3 4 5 6 7 8]]
9 10 11 12	[[1 2 3 4 5 6 7 8 9 10 11 12]]

Code Editor:

```
1 import numpy as np
2 rows,cols=map(int,input().split())
3 elements=[]
4 for _ in range(rows):
5     elements.extend(map(int,input().split()))
6 array=np.array(elements).reshape(rows,cols)
7 print(array)
8 print(array.ndim)
9 print(array.shape)
10 print(array.size)
11
```

Test Results:

Average time: 0.015 s, Maximum time: 0.031 s, 15.17 ms, 21.00 ms. 2 out of 2 shown test case(s) passed. 10 out of 10 hidden test case(s) passed.

Test case 1: Expected output: [[1 2 3 4 5], [6 7 8 9 10 11 12]]

Actual output: [[1 2 3 4 5], [6 7 8 9 10 11 12]]

Terminal: [1 2 3 4 5], [6 7 8 9 10 11 12]

Course

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Essentials of Data Science

Laboratory - 2304102L - 2026

Search course (ctrl + k)

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2. Practical 2

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2.1.1. List operations

2.1.2. Dictionary Operations

2.2. Lab Assignment

2.2.1. Linear search Technique

2.2.2. Captain of the Team

3. Practical 3

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3.1.1. Numpy Array Operations

3.2. Lab Assignment

3.2.1. Numpy: Matrix Operations

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

3.2.3. Numpy: Custom Sequence Generation

3.2.4. Numpy: Arithmetic and Statistical Operations

3.2.5. Numpy: Copying and Viewing Arrays

3.2.6. Numpy: Searching, Sorting, Counting

3.2.7. Student Data Analysis and Operations

3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

0.63s

stacking.py

1 import numpy as np

2

3 # Input matrices

4 print("Enter Array1:")

5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])

6

7 print("Enter Array2:")

8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])

9

10 # Perform horizontal stacking (hstack)

11 a=np.hstack((arr1,arr2))

12 print("Horizontal Stack:")

13 print(a)

14 # Perform vertical stacking (vstack)

15 b=np.vstack((arr1,arr2))

16 print("Vertical Stack:")

17 print(b)

Average time 0.036 s 36.25 ms

Maximum time 0.063 s 63.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Enter Array1:

1 2 3

4 5 6

7 8 9

Horizontal stack:

[[1 2 3 4 5 6]

[4 5 6 7 8 9]

Actual output

Enter Array1:

1 2 3

4 5 6

7 8 9

Enter Array2:

4 5 6

Terminal

Test cases

Prev Reset Submit Next

Course

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3.2.3. Numpy: Custom Sequence Generation

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using numpy based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

Test case 1

3

18

2

[3 5 7 9]

Test case 2

15

20

30

[18]

customS...

1 import numpy as np

2

3 # Take user input for the start, stop, and step of the sequence

4 start = int(input())

5 stop = int(input())

6 step = int(input())

7

8 # Generate the sequence using np.arange()

9 sequence=np.arange(start,stop,step)

10 # Print the generated sequence

11 print(sequence)

Average time 0.016 s 16.25 ms

Maximum time 0.021 s 21.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

Actual output

3 3

18 18

2 2

[3 5 7 9] [3 5 7 9]

Test case 2

Test case 3

Terminal Test cases

Prev Reset Submit Next

Course

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3.1.1. NumPy Array Operations

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3.2.2. NumPy Vector and Atrial Stack...

3.2.3. NumPy Custom Sequences Generator

3.2.4. NumPy Arithmetic and Statistical Op...

3.2.5. NumPy Copying and Viewing Array...

3.2.6. NumPy Searching, Sorting, Counting...

3.2.7. Student Data Analysis and Operations

4. Practical 4

5. Practical 5

2.2.4. NumPy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operators

You are given two arrays A and B. Your task is to complete the function array_operations, which will convert these lists into NumPy arrays and perform the following operations:

1. Arithmetic Operations:

- Compute the element-wise sum, difference, and product of the two arrays.

2. Statistical Operations:

- Calculate the mean, median, and standard deviation of array A.

3. Bitwise Operations:

- Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (i.e. A₁, (2), B₂).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

Test case 1

0 0 0 0

0 0 0 0

Element-wise Sum: 0 0 0 0

Element-wise Difference: -4 -4 -4 -4

Element-wise Product: 0 0 0 0

Mean of A: 0.5

Median of A: 0.5

Standard Deviation of A: 0.500000000000

Bitwise AND: 0 0 0 0

Bitwise OR: 0 0 0 0

Bitwise XOR: -4 -4 -4 -4

def array_operations(A, B):

1

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1000

Average time: 0.016 s

Maximum time: 0.024 s

1 out of 1 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output

0 0 0 0

0 0 0 0

0 0 0 0

Element-wise Sum: 0 0 0 0

Element-wise Difference: -4 -4 -4 -4

Element-wise Product: 0 0 0 0

Mean of A: 0.5

Median of A: 0.5

Standard Deviation of A: 0.500000000000

Actual output

0 0 0 0

0 0 0 0

0 0 0 0

Element-wise Sum: 0 0 0 0

Element-wise Difference: -4 -4 -4 -4

Element-wise Product: 0 0 0 0

Mean of A: 0.5

Median of A: 0.5

Standard Deviation of A: 0.500000000000

Terminal

Test cases

Course

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3.2.2 Numpy: Horizontal and Vertical Stack...

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3.2.5 Numpy: Copying and Viewing Arrays

3.2.6 Numpy: Searching, Sorting, Counting...

3.2.7 Student Data Analysis and Operations

4 Practical 4

5 Practical 5

3.2.5. Numpy: Copying and Viewing Arrays

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the original_array and assigning it to view_array.
- Creating a copy of the original_array and assigning it to copy_array.

After completing these steps, observe how modifying the view affects the original_array, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

original array after modifying view: <original_array>
View array: <view_array>

- After modifying the copy:

original array after modifying copy: <original_array>
Copy array: <copy_array>

Sample Test Cases

Test case 1

10 20 30 40 50 60 70 80

Original array after modifying view: [10 20 30 40 50 60 70 80]

View array: [10 20 30 40 50 60 70 80]

Original array after modifying copy: [10 20 30 40 50 60 70 80]

Copy array: [10 60 30 40 50 60 70 80]

Test case 2

90 2 3 4 5

Original array after modifying view: [90 -2 -3 -4 -5]

View array: [90 -2 -3 -4 -5]

Original array after modifying copy: [90 -2 -3 -4 -5]

Copy array: [90 88 -3 -4 -5]

copyAnd...

1 import numpy as np

2

3 inputlist = list(map(int,input().split(" ")))

4

5 # Original array

6 original_array = np.array(inputlist)

7

8 # Create a view

9 view_array = original_array.view()

10

11 # Create a copy

12 copy_array = original_array.copy()

13

14 # Modify the view

15 view_array[0] = 99

16 print("Original array after modifying view:", original_array)

17 print("View array:", view_array)

18

19 # Modify the copy

20 copy_array[1] = 88

21 print("Original array after modifying copy:", original_array)

22 print("Copy array:", copy_array)

23

Average time: 0.006 s

Maximum time: 0.013 s

7.75 ms

13.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

Expected output: 10 20 30 40 50 60 70 80

Actual output: 10 20 30 40 50 60 70 80

Original array after modifying view: [10 20 30 40 50 60 70 80]

Original array after modifying view: [10 20 30 40 50 60 70 80]

View array: [10 20 30 40 50 60 70 80]

View array: [10 20 30 40 50 60 70 80]

Original array after modifying copy: [10 20 30 40 50 60 70 80]

Original array after modifying copy: [10 20 30 40 50 60 70 80]

Copy array: [10 60 30 40 50 60 70 80]

Copy array: [10 60 30 40 50 60 70 80]

Terminal Test cases

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3.2.6: Numpy: Searching, Sorting, Counting

3.2.7: Student Data Analysis and Operations

4: Practical 4

5: Practical 5

3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

The given code in the editor takes a single array, array1, as space-separated integers as input from the user. Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

- Searching: Find the indices where search_value appears in array1 and print these indices.
- Counting: Count how many times count_value appears in array1 and print the count.
- Broadcasting: Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
- Sorting: Sort array1 in ascending order and print the sorted array.

Input Format:

- A single line containing space-separated integers representing array1.
- An integer search_value represents the value to search for in the array.
- An integer count_value represents the value to count in the array.
- An integer broadcast_value represents the value to add to each element of the array.

Output Format:

- The indices where search_value occurs in array1.
- The count of occurrences of count_value in array1.
- The array after adding the broadcast_value to each element.
- The sorted array.

Sample Test Cases

Test case 1

1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

[3 3 3 4 4 4]

[1 1 1 2 2 2]

arrayOpe...

1 import numpy as np

2

3 # Input array from the user

4 array1 = np.array(list(map(int, input().split())))

5

6 # Searching

7 search_value = int(input("Value to search: "))

8 count_value = int(input("Value to count: "))

9 broadcast_value = int(input("Value to add: "))

10 a=np.where(array1==search_value)[0]

11 print(a)

12 b=np.count_nonzero(array1==count_value)

13 print(b)

14 c=array1+broadcast_value

15 print(c)

16

17 d=np.sort(array1)

18 print(d)

19 # Count occurrences in array1

20

21 # Broadcasting addition

22

23 # Sort the first array

Average time 0.019 s 19.00 ms

Maximum time 0.024 s 24.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1 6/10ms Debug

Expected output 1 1 1 2 2 2

Actual output 1 1 1 2 2 2

Value to search: 1

Value to count: 2

Value to add: 2

[0 1 2]

3

[3 3 3 4 4 4]

[1 1 1 2 2 2]

Terminal Test cases

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